

TechStart Azure Security & Modernisation Project

Business Scenario

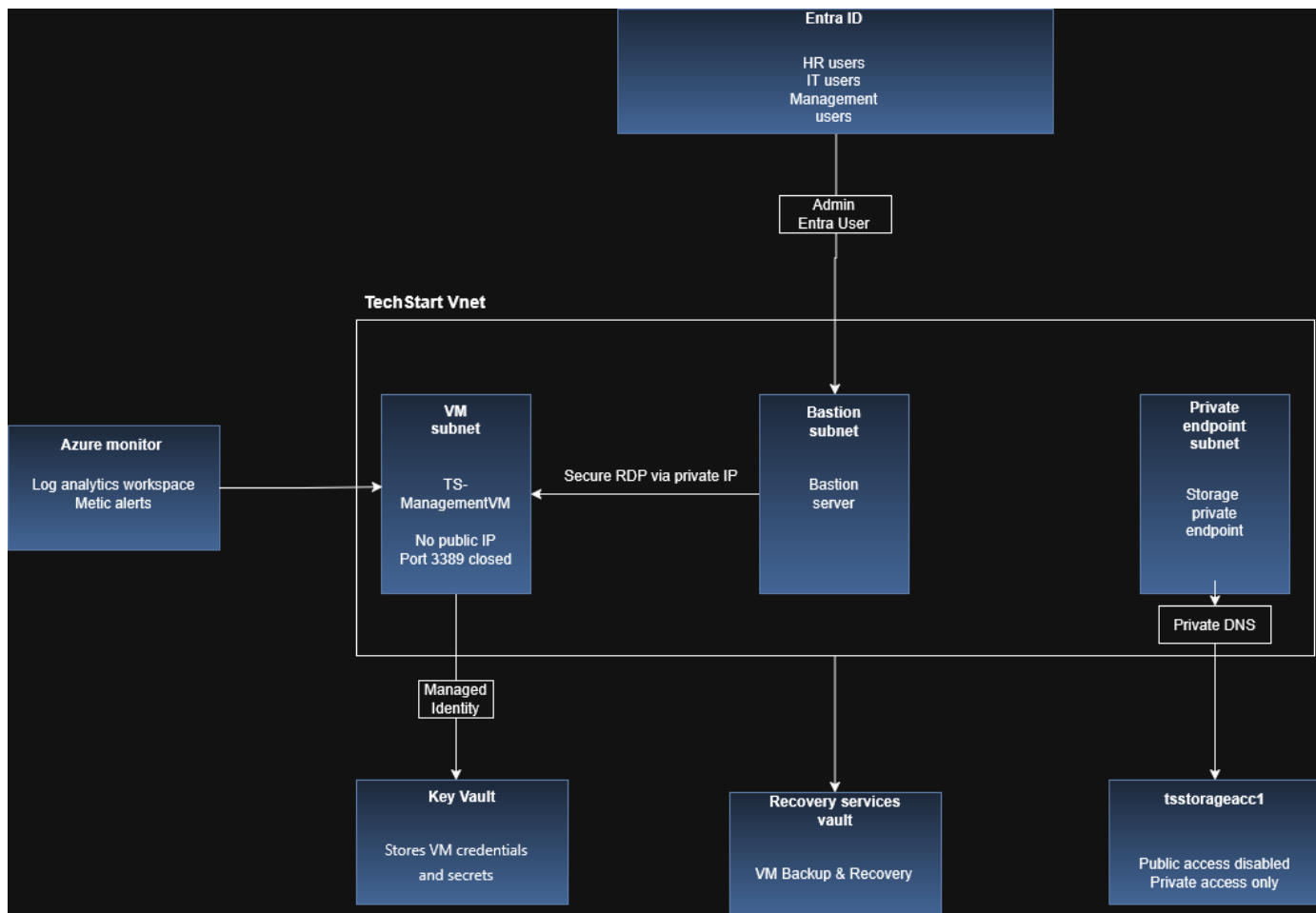
TechStart Ltd is a rapidly growing technology startup with 15 employees across HR, IT and Management departments.

Following an internal infrastructure review, several operational and security risks were identified:

- Sensitive HR records were stored in unsecured shared locations.
- Administrative passwords were being stored in spreadsheets and shared documents.
- No formal backup or recovery process existed for critical systems.
- Administrative access methods did not align with security best practices.
- Administrative tasks were performed from unmanaged endpoints, creating security and auditing concerns. A centralised management server was required to provide secure administration of cloud resources and controlled access to business data.
- Monitoring and alerting capabilities were limited.
- The organisation planned to adopt Microsoft Azure but had no defined cost governance or resource governance standards.
- A monthly cloud budget of £50 has been allocated for the initial deployment.
- The company operates exclusively within the United States, requiring all resources and data to remain within approved US regions.

The business required a secure Azure environment capable of supporting current operations while providing a scalable foundation for future growth. The solution needed to improve security, centralise administrative access, protect sensitive business data, and establish monitoring, backup and recovery capabilities aligned with industry best practices.

Architecture Diagram



Technical Design Summary

Identity & Access

- Entra ID users and groups created.
- RBAC implemented using least privilege.

Reason

Provide controlled access based on department responsibilities.

Secure Administration

- Windows Server 2025 VM deployed.

- Azure Bastion implemented.
- Public RDP removed.

Reason

Reduce exposure to internet-facing administration services.

Credential Management

- Azure Key Vault deployed.
- VM administrator credentials secured.
- Managed Identity enabled.

Reason

Centralise privileged credential management and reduce credential exposure.

Storage Security

- Azure Storage Account deployed.
- HR and IT containers created.
- Private Endpoint and Private DNS configured.
- Public Storage access disabled.

Reason

Protect sensitive business data using private Azure networking.

Backup & Recovery

- Recovery Services Vault deployed.
- Daily VM backups configured.
- Restore testing completed.

Reason

Provide recovery capability and support business continuity.

Monitoring

- Log Analytics Workspace deployed.
- Azure Monitor alerts configured.

Reason

Provide operational visibility and proactive alerting.

Cost Management & Governance

- Azure Budget configured.
- Azure Policy implemented.

Reason

Control spending and enforce deployment standards.

Build Evidence

Entra ID

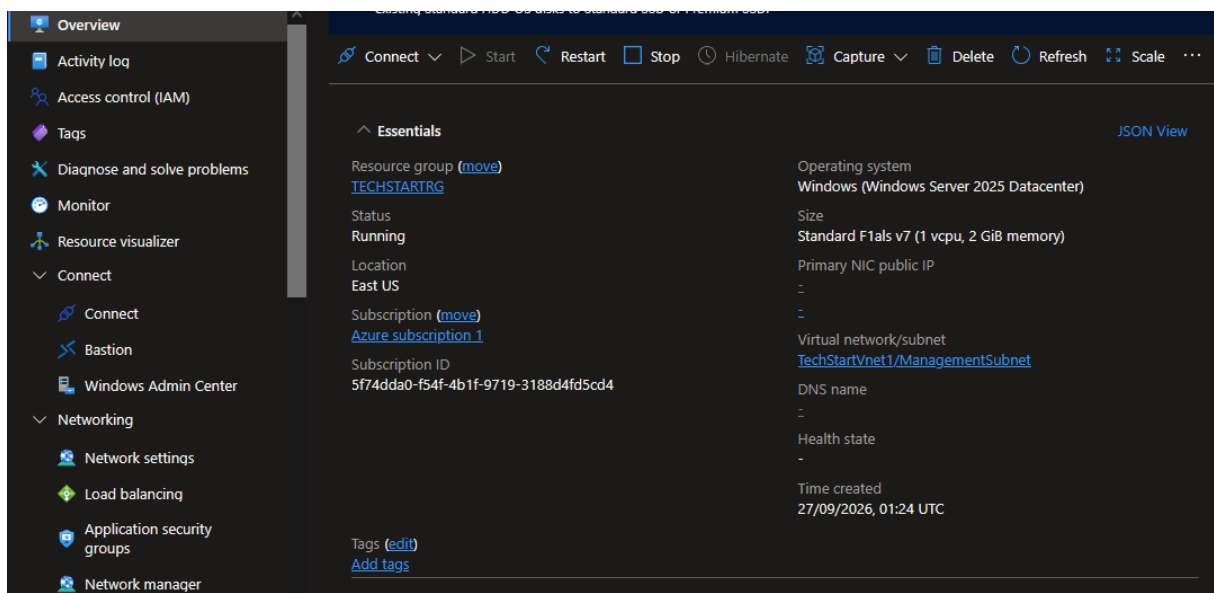
- I created the user accounts and security groups for the three departments; users are added to the security groups based on which department they work in.

☐	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sy...	Sign-in identifiers	Company name	Creation type
☐	AA Aaliyah Akhtar	aaliyahaakht...	Member	No	No	View		
☐	AM Ali Malik	Ali.Malik@a...	Member	No	No	View	TechStart LTD	
☐	DW David Walker	David.Walke...	Member	No	No	View	TechStart LTD	
☐	EB Emma Brown	Emma.Brow...	Member	No	No	View	TechStart LTD	
☐	JS James Smith	James.Smith...	Member	No	No	View	TechStart LTD	
☐	KW Kate Wilson	Kate.Wilson...	Member	No	No	View	TechStart LTD	
☐	SI Sarah Jones	Sarah.Jones...	Member	No	No	View	TechStart LTD	
☐	TH Tom Hughes	Tom.Hughes...	Member	No	No	View	TechStart LTD	

☐	Name ↑	Object id	Group type	Membership type	Email
☐	H HR-Group	94ea28f6-9f68-43c3-b985-6e96d65beeaf	Security	Assigned	
☐	I IT-Group	d3820983-b1d9-471e-8614-a8d2fac441db	Security	Assigned	
☐	M Management-Team	4568ab52-0ebf-45d6-b875-f65a115ee7b3	Security	Assigned	

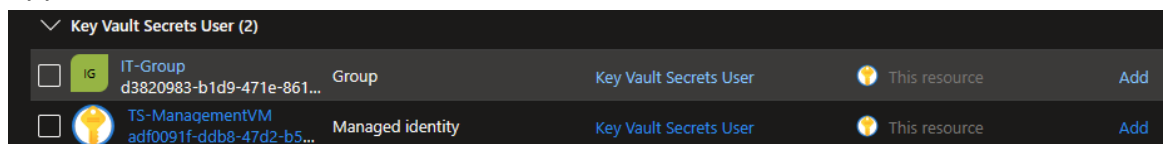
Virtual machine deployment

- VM name - TS-ManagementVM, image (Windows Server 2025 Datacenter)
- Standard F1als v7 (1 vcpu, 2 GiB memory)
- No public IP was created and assigned, RDP port 3389 is closed as access to the VM will be done through the Azure portal and Bastion server for secure access
- A dedicated data disk was attached to separate operating system data from business data and support future expansion requirements.
- NSG attached to the VM left on default settings.
- TS-ManagementVM was deployed as a dedicated management server used by the IT team for administration, monitoring, recovery operations and access to protected Azure resources.



Azure key vault

- The local administrator credentials were stored as secrets in Azure Key Vault.
- Key vault access was secured using RBAC, the IT group was granted key vault secret user and the VM was setup as a managed identity allowing RBAC to be applied to the VM.



- This allows centralised management of secretes and sensitive information, removing the dependency on using unsecure locations to store passwords and secrets.

Storage account

- A blob storage account was configured and blob containers added, RBAC was implemented at the container level.
- The storage account was secured with a private endpoint and disabling public network access, which protects sensitive data.
- Private endpoint access tested from within the VM via nslookup.

The screenshot displays the Azure portal interface for a storage account. At the top, a toggle for 'Public network access' is shown as 'Disabled' with a 'Manage' button. Below this is a table of private endpoints:

☒	Name	Connection name	Connection state	Description	
☐	PrivateEndpointSA	...	tsstorageacc1.c4b38111-8557-4e...	Approved	Auto-Approved

Below the table is a detailed view of the private endpoint configuration:

Configuration name	FQDN	IP address	Subscription	Private DNS zone	DNS zone group
▼ privatelink-blob-core-w...			Azure subscription 1	privatelink.blob.core.windows.net	default
▼	tsstorageacc1.privatelink.blob.core.windows...	10.0.2.4		-	-

- Private DNS resolves the storage account host name to the private endpoint IP, so this confirms the private endpoint is configured and working.

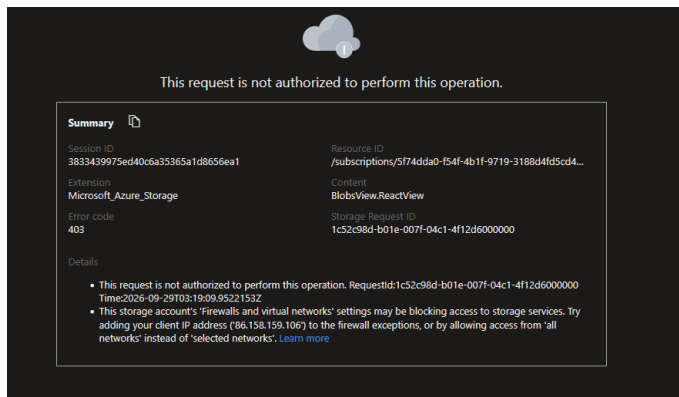
```
Administrator: Windows Powe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\LocalAdmin> nslookup tsstorageacc1.blob.core.windows.net
Server: UnKnown
Address: 168.63.129.16

Non-authoritative answer:
Name: tsstorageacc1.privatelink.blob.core.windows.net
Address: 10.0.2.4
Aliases: tsstorageacc1.blob.core.windows.net

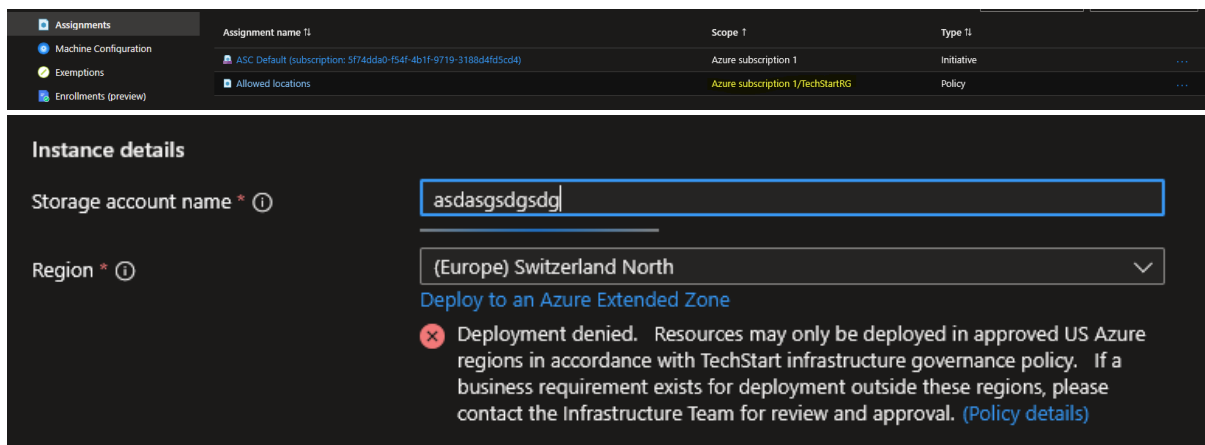
PS C:\Users\LocalAdmin>
```

- Storage security validation, I attempted to access Azure Storage containers from an external client outside the TechStart virtual network. Expected Result Access denied because Public Network Access was disabled and the Storage Account was secured using a Private Endpoint.

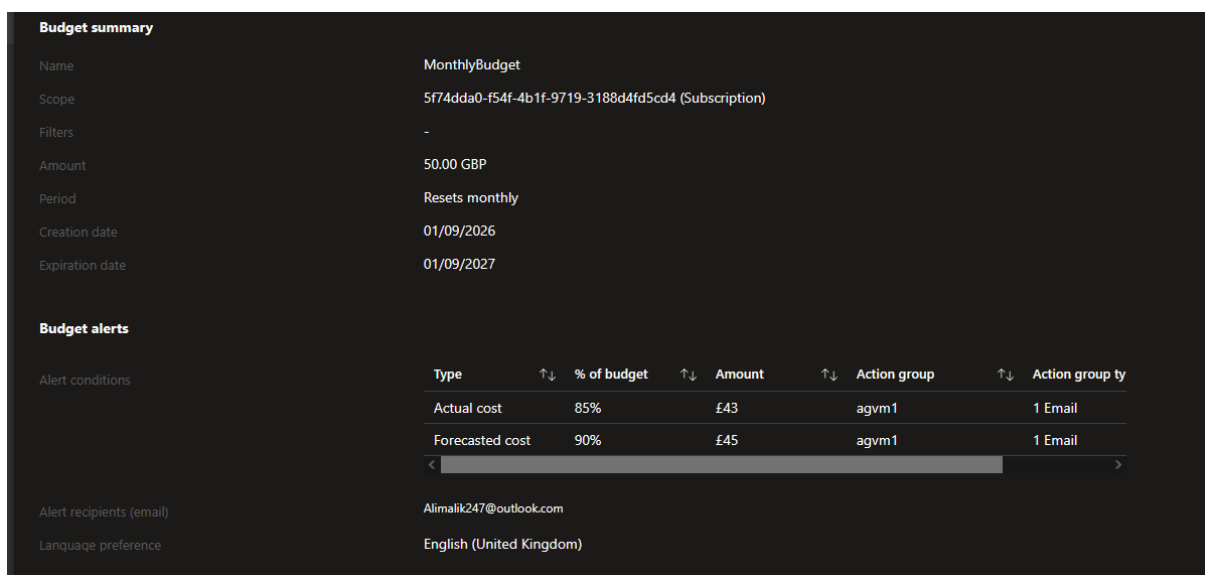


Azure policy and Governance

- 'Allowed locations' policy was set at the resource group level allowing deployments in US regions only, a custom message was applied and tested



- As the business scenario states a £50 monthly spend budget has been provided as the Cost and governance controls implemented. Alerts were at forecasted and actual cost levels, see screen shot.



Backup/ Recovery services vault

- Virtual machine backup policy (enhanced), VM restore tested and deployed

Restore Virtual Machine ...

TS-ManagementVM

Restore allows you to restore VM/disks from a selected Restore Point.

Restore point *
[Select](#)

Data Store

Restore configuration

Restore target ☒ Create new
☐ Replace existing

To create an alternate configuration when restoring your VM (from the following menus), use PowerShell cmdlets.

Restore Type * ☐

Virtual machine name * ☐

Availability Zone * ☐

Restores within the same zone and subscription will utilize Snapshot tier. For restores to a no zone, different zone or subscription, Vault tier restore points will be used.

Subscription * ☐

Resource group * ☐

Virtual network * ☐

Subnet * ☐

Staging Location * ☐
[Can't find your storage account?](#)

Restore

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

Disks + images

Subscription equals all

Type equals all

Resource Group equals all

Location equals all

+ Add filter

☐	Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size
☐	ManagementVmTestRestore	Azure subscript...	TechStartRG	East US	Running	Windows	Stand...
☐	TS-ManagementVM	Azure subscript...	TECHSTARTRG	East US	Stopped (deall...	Windows	Stand...

- Also Blob storage soft delete, container soft delete and blob versioning has been configured to allow recovery of delete files, container and versioning allows recovery of document versions. Infrastructure encryption was enabled to provide an additional layer of protection for the sensitive data in the storage account.

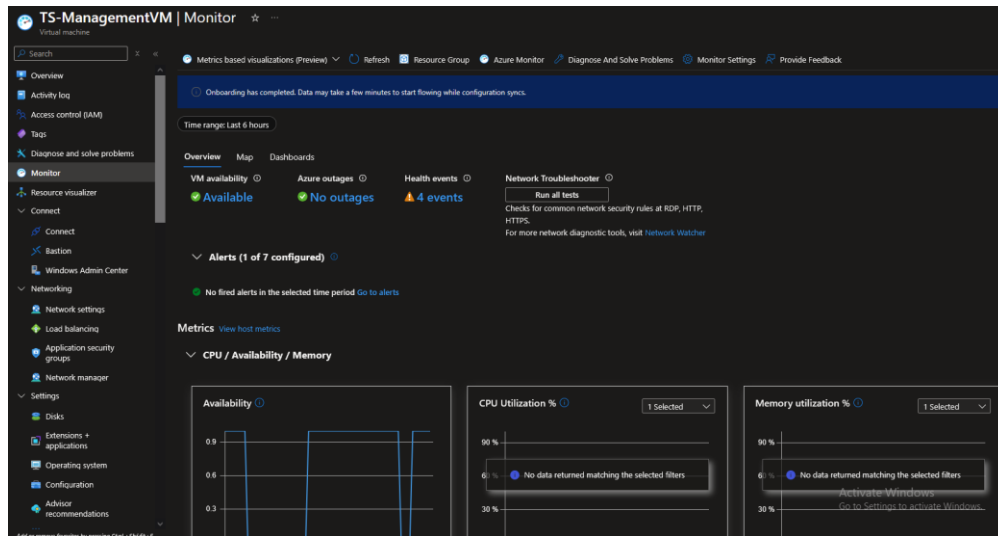
Blob soft delete
Enabled (30 days)

Container soft delete
Enabled (30 days)

Versioning Enabled

Log analytics workspace

- Log Analytics Workspace deployed and connected to TS-ManagementVM.
- Azure Monitor configured for VM availability and performance monitoring.



Validation & Testing

- ✓ Azure Bastion successfully used to access TS-ManagementVM without public RDP exposure.
- ✓ Administrator credentials securely stored in Azure Key Vault.
- ✓ Managed Identity configured and assigned appropriate RBAC permissions.
- ✓ Department-based RBAC validated for HR, IT and Management teams.
- ✓ Storage Account secured using Private Endpoint and Private DNS.
- ✓ Public Storage Account access disabled and validated.
- ✓ DNS resolution confirmed Storage Account traffic is routed through the Private Endpoint.
- ✓ Recovery Services Vault backup completed successfully.
- ✓ Virtual machine restore completed successfully.
- ✓ Azure Monitor alerts configured for VM availability and performance monitoring.
- ✓ Azure Budget and cost alerts configured successfully.

- ✅ Azure Policy validated and restricting deployments to approved regions.

Outcome

The environment was successfully deployed, tested and validated. Security, backup, monitoring, governance and access control requirements were achieved and verified through testing.

This project provided hands-on experience with designing, deploying, securing and validating Azure infrastructure using industry best practices.

Things that broke and key lessons learnt

- For Azure Monitor to begin collecting operating system telemetry from a virtual machine, the Azure Monitor Agent must first be installed and configured on the VM.
- Configuring VM backups was straightforward; however, completing a successful restore test was an important validation step and provided confidence that recovery procedures would work during a real outage.
- Managing access through security groups rather than assigning permissions directly to individual users proved significantly easier to administer and maintain.
- Azure Bastion provided secure remote administration of the management server without the need to expose RDP over a public IP address.
- Disabling public network access on the storage account and implementing a private endpoint ensured that storage could only be accessed through the private network, removing direct access from the public internet.

Summary

Designed and deployed a secure Azure infrastructure environment incorporating network isolation, identity-based access control, backup and monitoring services. The solution demonstrates core cloud infrastructure principles including security, governance, operational monitoring, business continuity and private connectivity.